

From Credit Risk Modelling to Credit Policy

A data-driven case study using a German mortgage portfolio

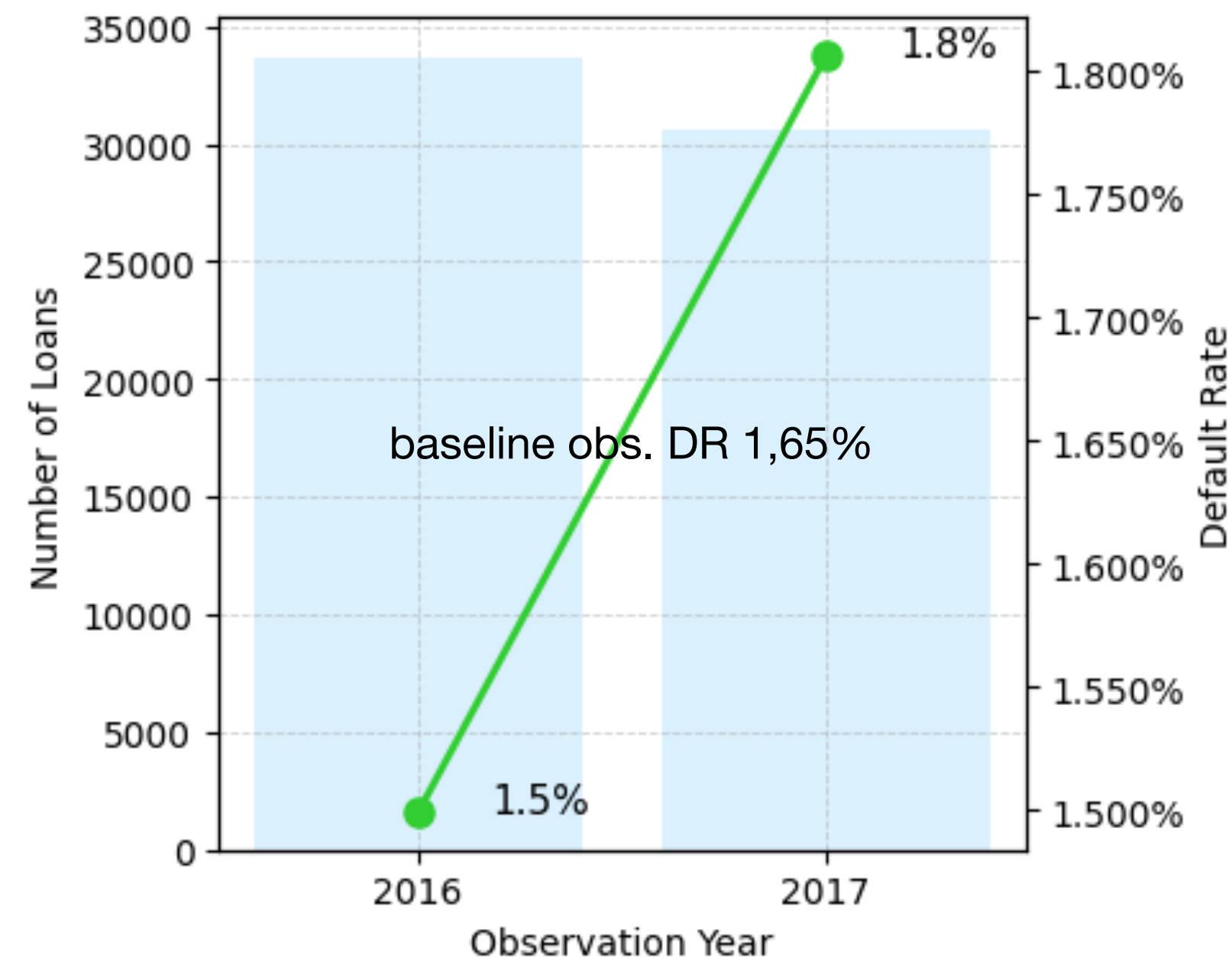
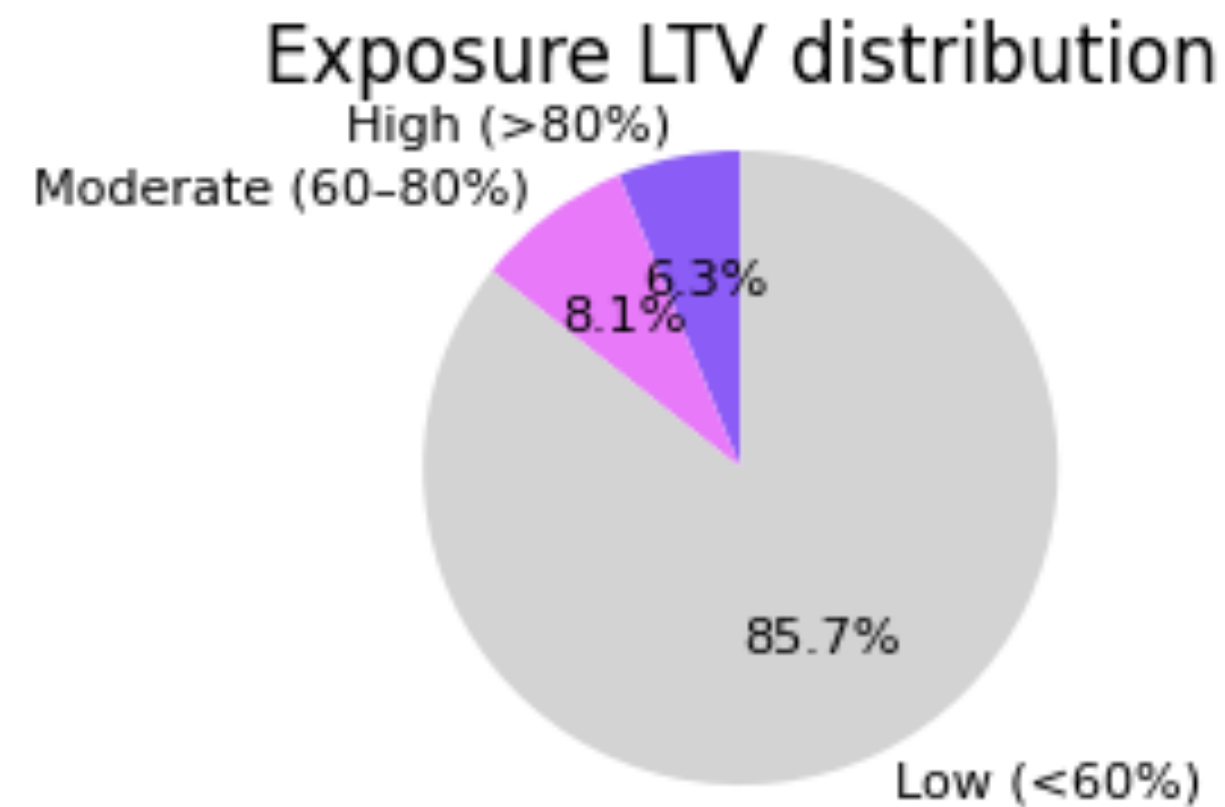
Understanding the Mortgage Portfolio

Objective: Differentiate mortgage default risk and translate PD estimates into credit-policy decisions.

Dataset: ~64k German 'General Mortgages' observations, 2016–2017.

Approach: 1. Compare structural vs behavioural risk information.

2. Evaluate PD, LTV, and combined credit policies.



Behavioural Signals improve Risk differentiation

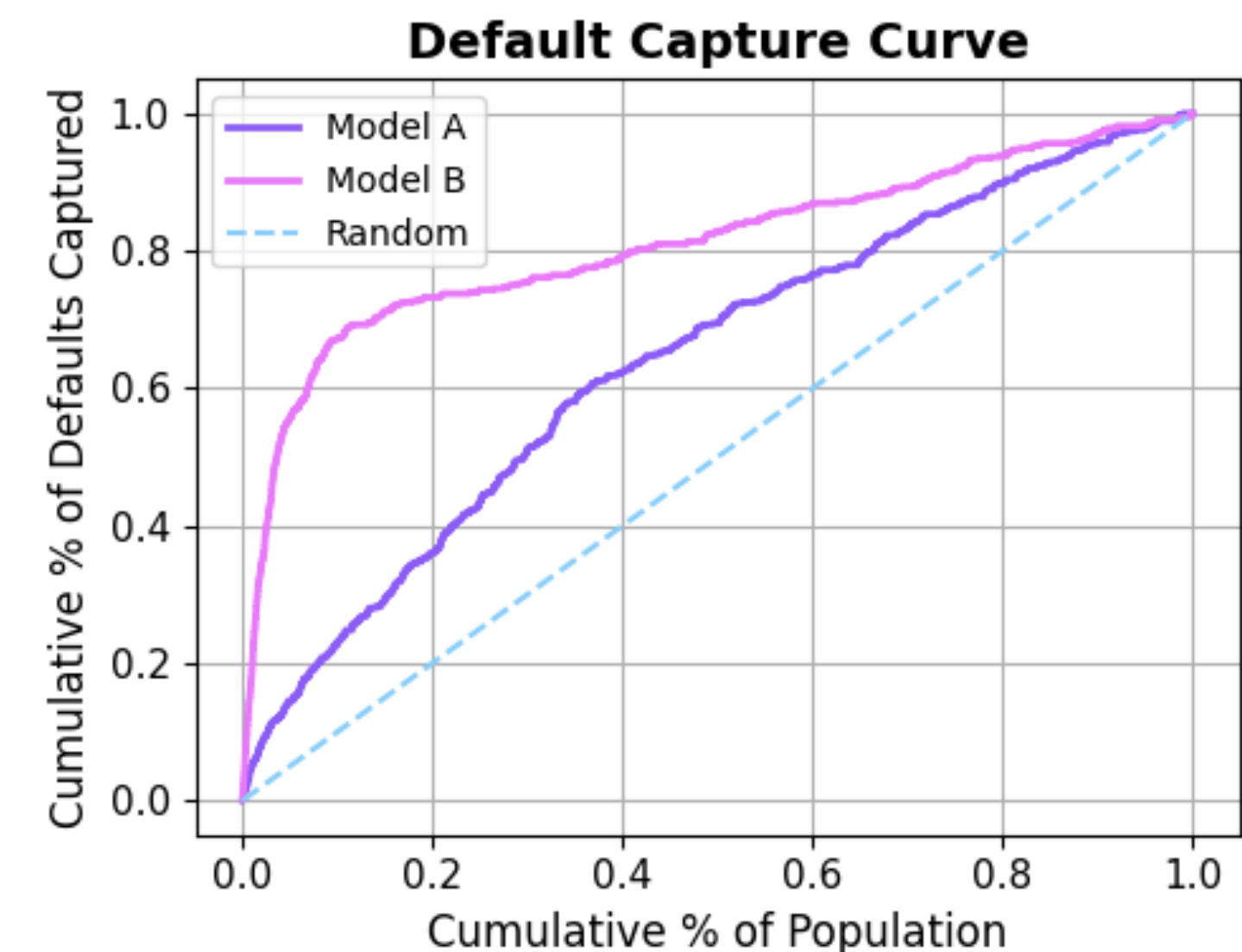
Adding behavioural information substantially strengthens out-of-time risk differentiation.

Model B is therefore used as the PD model for policy analysis.

Key predictive signals:

- ▲ 30+ Days Delinquent (last 12 months)
- ▲ Days in Delinquency

	Model A	Model B
Features number	12	22
Features type	Structural	Structural + Behavioural
AUC (OOT)	0,64	0,82
Obs DR	1.81%	1.81%
Avg PD	1.32%	1.52%
Brier	0,018	0,015



Policy Evaluation

	no policy	PD Policy	LTV Policy	Combined Policy
Policy rule	-	PD<0,20	LTV<80%	PD<0,20 + LTV<80%
Exposure (€bn)	15,2	14,8	13,1	12,8
Exposure retained (%)	100	97	87	84
Obs DR (%)	1,81	1,24	1,73	1,22
Avg PD (%)	1,52	0,96	1,43	0,93
Monte-Carlo EL (€m)	126,3	69,1	96,9	56,9
Economic Capital (€m)	19,6	16,2	18,0	14,1

*Policy evaluation performed on the 2017 OOT population. Monte Carlo EL and Economic Capital assume LGD = 40%.

Combining PD and LTV constraints reduces observed risk, expected loss and economic capital beyond either policy individually.